Conflicts, technologies and global security

On the use of deepfakes in modern information warfare

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Abstract

Deepfakes, synthetic audio or visual content generated by artificial intelligence models, have become one of the most powerful weapons of contemporary information warfare. This report first gives a *geostrategic* state of the art of their use, through a series of recent conflicts (Russia–Ukraine, Israel–Hamas, India–Pakistan, Syria, sub-Saharan Africa), then a *technical* state of the art of their generation (GANs, diffusion models, voice synthesis), of their detection and of the countermeasures currently in place. On this basis, it puts forward a set of solutions organised around three levers, *detect* (participatory CAPTCHAs, blockchain traceability), *penalise* (a non-proliferation framework, generative AI licences) and *educate* (media literacy, active learning), before examining the shift towards an *offensive* use of deepfakes by states. The central thesis is that the fight must become a systemic, permanent and adaptive effort, combining technological innovation, citizen participation and international cooperation.

1 Introduction

Deepfakes are synthetic audio or visual content generated by artificial intelligence models capable of imitating a voice, a face or a movement in a way that is indistinguishable from the original. Built on neural networks, deepfakes are today one of the most powerful and most widespread techniques of information manipulation. Their massive circulation, facilitated by social networks and by the proliferation of tools available to the general public, raises major issues for international security, public trust and geopolitical stability, and is paradigmatic of a tightening link between technology and modern conflicts.

After setting out a state of the art of the current use of deepfakes in information warfare, we will formulate several recommendations, applicable at different scales, in order to respond effectively to this growing threat. Our report will therefore seek to answer the following question: how should states and institutions approach the use of deepfakes in a context of constant information and influence warfare?

2 State of the art of the use of deepfakes

2.1 Geostrategic state of the art

Deepfakes are today a strategic tool deployed in many conflicts. Their use most often serves two major intentions: influencing public opinion and destabilising the adversary strategically. In order to understand the specific features of the use of deepfakes as a "weapon of war", we examine a few recent cases below.

2.1.1 Russia – Ukraine

The Russo-Ukrainian conflict is one of the most documented settings for the use of deepfakes (Singh, 2025). In March 2022, a deepfake of Volodymyr Zelensky circulated, imitating him calling on Ukrainian soldiers to lay down their weapons. A symmetric version targeted Vladimir Putin, announcing a fake general mobilisation. Although the quality of these deepfakes was fairly crude (they can very easily be identified as such), their psychological objective was clear: to introduce doubt, to fracture mental cohesion, to test the porosity of information (Barthélemy, 2025). Their rapid circulation underlines the strategic importance (Majchrzak, 2023) of digital platforms in hybrid influence operations, in particular Telegram and X (formerly Twitter), which have been flooded with such examples, and which intelligence services often infiltrate in order to determine how to use them as vectors for spreading false information.

2.1.2 Israel – Hamas

In the Israel-Hamas conflict, deepfakes of wounded children or of civilians in distress were massively circulated in order to polarise public opinion (Thomson, 2023; Kazic, 2024). Visual fabrications, some of them fairly crude, others far harder to identify, circulated faster than the official denials. This content is part of a strategy aimed at maximising emotional impact in order to mobilise or radicalise the public, to amplify indignation or anger, and to saturate the media space. It is worth noting that the main target was international rather than local public opinion, the aim being to attract the sympathy of potential allies. It is also worth noting that this is a tool that sits among a whole battery of others within a global communication strategy: on the Israeli side, it falls within *hasbara*, which covers every use of communication strategies as part of public diplomacy.

2.1.3 India – Pakistan

The Indian subcontinent is the scene of intense disinformation activity. In this context, deepfakes have been used to simulate Pakistani soldiers acknowledging a defeat, or to

imitate Indian religious leaders calling for revenge (Singh, 2025). This content is amplified by communal tensions and by encrypted messaging groups, where it circulates without any control. Its effectiveness rests on its capacity to exploit identity fault lines, and it heightens the risk of escalation. It also introduces an uncertainty that complicates strategic analysis (Kashif-Janjua, 2025): a message attributed to an official actor can now be perceived as potentially fake, which weakens institutional trust.

2.1.4 Syria

When the regime of Bashar al-Assad fell, a deepfake of the leader announced his refusal to capitulate and called on loyalist forces to continue the fight. Distributed in areas where official information circulated poorly, this deepfake created considerable confusion (Kazic, 2024), illustrating the vulnerability of populations in contexts of media fragmentation. The strategy here is characteristic of psychological warfare aimed at disrupting the chain of command (Byman et al., 2023). A falsified announcement of surrender or, on the contrary, of fierce resistance can alter tactical behaviour on the ground. This case illustrates the specific danger of deepfakes in asymmetric warfare environments, where communications are sporadic and where a well-placed synthetic message can have a disproportionate effect.

2.1.5 Sub-Saharan Africa

In several regions of sub-Saharan Africa, notably in Ethiopia, South Sudan and certain Sahelian areas, voice deepfakes imitating community leaders have encouraged reprisals or announced fake imminent attacks (Breland, 2019). In societies where the traditional media are weak and where community radio plays a central role, the spread of audio deepfakes constitutes a serious threat (Johnson, 2019). These deepfakes are often combined with pre-existing rumours, producing an effect of informational resonance. Interethnic violence can thus be triggered or amplified by a single falsified voice message. The impact therefore goes beyond disinformation: this is a manipulation tool with a potentially lethal capacity.

2.1.6 Conclusions

The analysis of all these cases reveals one constant: the effectiveness of deepfakes rests not only on their realism, but above all on their capacity to infiltrate environments that are already unstable, emotional or polarised. They are first and foremost instruments of destabilisation, used to disturb perception, weaken authority, fragment societies and alter strategic balances. They are also, naturally, aimed mainly at civilian populations, in order to influence public opinion.

While this report was being written, on 16 December 2025, a deepfake purporting to show a coup in France was shared widely on social networks (12M views), notably on Facebook. The impact of this deepfake, which showed a journalist on the banks of the Seine, was such that an African president (whose identity was not revealed) contacted Emmanuel Macron. Apparently produced with no intent to harm, this deepfake nonetheless underlines how easily such situations can escalate.

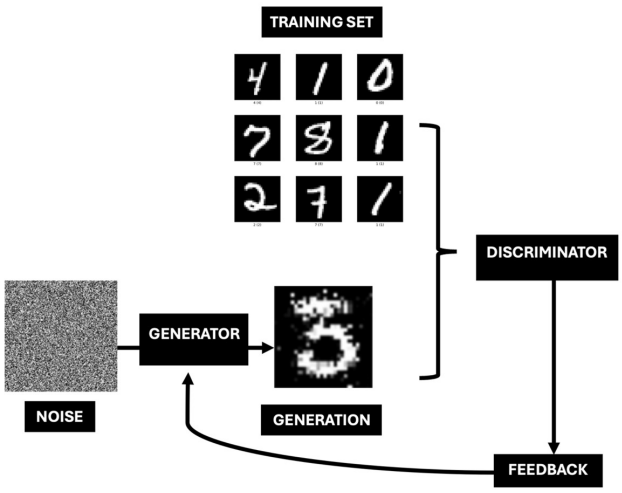


Figure 1. General operation of a GAN (*generative adversarial network*).

3 Technical state of the art

3.1 Generating deepfakes

The generation of deepfakes relies on several families of models (Owasp, 2023):

- **GANs** (*generative adversarial networks*): a generator, based on artificial intelligence, produces realistic images which it submits to a discriminator (also an AI) that tries to distinguish them from real images, creating a competition that gradually improves their quality, until the discriminator judges the image satisfactory (Bonnet et al., 2025);
- **Diffusion models** (Bonnet et al., 2025): they generate images starting from "noise", that is, random pixels, and progressively refining the composition of the image from a training database;
- **Voice models** (Bendiab et al., 2025): from language models and the learning of spectral signatures, they make it possible to generate speech and particular voices.

These techniques are now available to the general public, which considerably lowers the barriers to entry for non-state actors (such as terrorist groups or local militias): platforms such as DeepFaceLab and FaceSwap, or automated services such as HeyGen and DeepfakeWeb, make it possible to create deepfakes in a matter of seconds.

3.2 Detecting deepfakes

Deepfake detection rests on two main approaches. The first is algorithmic (Cinar, 2023): supervised models learn to identify visual anomalies (facial expressions, shadows, shapes), audio ones (spectral phase shift, absence of coherent background noise) or multimodal ones (lip desynchronisation) (Abdulhamied et al., 2025). The second is forensic: metadata verification, traceability analysis through digital signatures and file provenance.

However, the balance between generation and detection remains unstable: generative models progress faster than detectors. As in any adversarial competition, a purely algorithmic approach is bound to fail in the long run, since it feeds the very networks it is trying to fight.

3.3 Countering deepfakes

Current strategies for countering deepfakes fall into three main areas: regulation, detection and public education.

On the **legal side**, several states have adopted frameworks aimed at making platforms accountable, at imposing rapid takedown obligations or at sanctioning the malicious manipulation of information. The United States, for instance, has introduced state-level election laws prohibiting the distribution of political deepfakes close to a vote, while the European Union is strengthening traceability and reporting standards through the DSA (*Digital Services Act*). In France, a legal framework has existed since 2024: the penalties go "only" up to a fine of €45,000 and 2 years in prison for distributing deepfakes of a non-consenting person that are not labelled as AI-generated, and up to a fine of €75,000 and 3 years in prison for pornographic content (French Government, 2024). Case law on the creation or distribution of deepfakes, for its part, is still being built. Thus, in January 2026, the French government referred the matter to the public prosecutor for the first time in order to open an investigation into the social network X because of its AI *Grok*, massively used by internet users to undress women, sometimes minors. Unfortunately, in practice only users face the penalties set out above, and Elon Musk's main reaction was to put the image-generation tool behind a paywall rather than to consider any moderation of it (Deromas-Lebocq, 2026).

On the **technical side**, platforms are experimenting with *watermarking* or digital marking systems intended to distinguish AI-generated content, complemented by automated detection tools based on forensic analysis (image inconsistencies and defects).

On the **educational side**, some awareness campaigns seek to teach citizens to spot the signs of visual manipulation, to check sources and to adopt a critical stance towards viral images: examples include the podcast "Les Voies de l'IA" offered by the Observatoire de l'IA, or the KlickSafe initiatives at the European level, such as the webinar "Deep Fake, Deep Impact" on 1 October 2025 and the provision of teaching material against cyber-manipulation.

These approaches, however, remain partial and heterogeneous. Regulation advances slowly and remains dependent on an inter-state consensus, whereas disinformation circulates globally. Technical mechanisms, useful as they are, are fragile in the face of the rapid improvement of generative models. As for educational initiatives, they struggle to reach the most vulnerable audiences or the conflict zones, where deepfakes are precisely the most destabilising. The current geopolitical reality therefore calls for a more integrated strategy, capable of combining technological innovation, citizen participation, international cooperation and the anticipation of the new offensive capabilities offered by AI, simultaneously. In other words, the fight against deepfakes must become a systemic, permanent and adaptive effort, rather than a set of isolated and reactive initiatives.

4 Proposed solutions

4.1 Detect

4.1.1 CAPTCHAs to strengthen detection and resilience

A purely technical solution would consist in turning the classical concept of the CAPTCHA into a tool that trains humans and detection models at the same time. Instead of identifying traffic lights or pedestrian crossings, the user would be faced with visual or audio elements containing deepfakes. The system pursues a twofold objective:

1. to strengthen the human capacity to recognise deepfakes, through repeated exposure to deepfakes used in real situations;

2. to feed the detection models, thanks to the massive and distributed participation of users.

This approach is highly scalable: millions of internet users can take part without any additional effort, as they already do for traditional CAPTCHAs. It also reaches every age group, including adults and elderly people, unlike digital education campaigns that address young people only. Next, it provides continuous improvement of the models thanks to a constant flow of annotations: detection is no longer based solely on one recurring anomaly or another, but capitalises on the human capacity to tell real content apart from artificial content. It also strengthens the cognitive resilience of populations faced with visual manipulation, fosters a form of collective vigilance and integrates easily into existing platforms. Finally, it is a low-cost strategy, resting on infrastructure that is already deployed.

Of course, this solution as it stands is not without limitations: above all, it rests on trust in the human capacity to discriminate fake content, and if that capacity is overestimated, we run the risk of degrading the algorithmic detection models that we feed with the answers of internet users. To mitigate this risk, the deployment of this solution should be preceded by a testing phase, in order to estimate the human detection capacity. One could also consider adopting a "pedagogical" mode that offers, after the CAPTCHA, brief feedback highlighting the details that should have alerted the user. Another limitation concerns the geographical effectiveness of the solution: the populations most exposed to deepfakes are also the least well informed, and their use of the internet is probably lower than in developed countries. To address this problem, one could consider cloning the CAPTCHAs onto other information channels: as games in newspapers, on the radio, or on television. This of course presupposes the participation of local authorities, but insofar as they are often themselves threatened by the abundance of deepfakes, this point does not appear to be the main limiting factor.

4.1.2 Blockchain to improve content traceability

One possible solution for fighting deepfakes would be to use a recent but reliable and effective technology: the blockchain (Rashid, 2021). This is a database allowing several individuals to share and store information, and to do so anonymously, securely and without going through

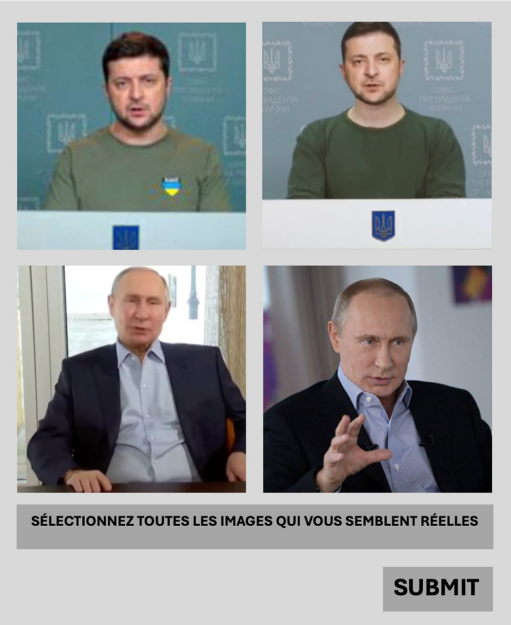


Figure 2. Example of an anti-deepfake CAPTCHA that could be proposed.

intermediary platforms. The point is therefore that when a piece of information is added or modified, everything is recorded in the blockchain through a *hash*, that is, a digital identifier. It is thus possible to trace the life of the data. Concretely, timestamped proofs are stored there (transactions, certificates, file fingerprints): each entry contains a unique identifier, the digital signature of the issuer and a date, which makes it possible to verify the origin and the integrity of a file without going through a trusted intermediary. The strength of the blockchain therefore comes from its decentralisation, from cryptography and from transparency (Freeman, 2025).

Consequently, in the case of deepfakes, once the original video has been recorded on the blockchain, one could tell, when looking at a modified version of the original, that it is a fabrication. Concretely, to put this in place, one would have to create a blockchain shared by all media outlets, journalists, politicians, influencers and, more broadly, everyone likely to publish videos on information platforms. From then on, when a journalist films a video and records it on the blockchain, a *hash* will be assigned to that video. Thus, when a malicious individual wishes to publish an edited version of that video, it will be enough to look at its *hash*, then to check in the blockchain database whether the *hash* corresponds to that of a video recorded by a verified entity. If the *hash* does not correspond to any other video, it means that this is not an official video, and therefore, if it shows public figures, that it has probably been faked.

Thus, through a simple check, it would be possible to know whether a video is official or not. This verification could be carried out in various ways, more or less effective. It could be done through a partnership with the platforms (Patil, 2021): it would be their role to carry out this verification, in which case they could indicate directly to users that the content is a fabrication. Otherwise, it could be done by the users themselves, thanks to a dedicated piece of software created by the government. This last method would necessarily be less effective since it would require the user to carry out the check, and therefore to have installed the software beforehand and to know how to use it. One can thus imagine an international cooperation to create a blockchain shared by different countries, which could cover a large share of published content.

Nevertheless, this solution has limits. First, if an entity wishes to spread a fake video that does not reuse any other already produced video, this cannot be detected. Typically, deepfakes of wounded civilians in the Israeli-Palestinian conflict cannot be detected as fabrications by the blockchain, because there is no information about the original video; at best, we will know that it was not published by official media. Moreover, a shared blockchain created by a government may lead some media outlets to fear censorship or manipulation by state bodies, and therefore to refuse to join the blockchain (Lokhande, 2024).

4.2 Penalise

Finally, it is worth considering what guarantees the law can offer to counter attacks whose legal remedies are complex, in a more global context than the agreements that already exist here and there in the world.

4.2.1 Curbing the proliferation of deepfakes

A first solution against the creation of deceptive content would be to adopt an international framework for the non-proliferation of deepfakes, along the lines of the Oslo (2008) and Ottawa (1997) treaties, which prohibit the production, possession, transfer and use of cluster munitions and anti-personnel mines respectively. The signature of these treaties by a country requires it, among other things, to declare its stockpiles to the secretariat of the Geneva Convention and to destroy them within a granted deadline (United Nations, 2026). Unfortunately, these treaties are only ratified by willing countries and it is entirely possible for them to withdraw. The Ukrainian battlefield is the perfect example of a violation of these rules, since use by both sides has been reported since 2022. This framework, ethically sound though it is, could therefore turn out to be more a source of imbalance between the powers that have the capacity to use deepfakes than anything else.

Another conceivable obstacle would be to limit the computing capacity of generative AI according to its use. One can imagine the creation of a categorisation of AI systems, each with a “licence” required for its use, whose number would be inserted as a watermark in the metadata of the generated images. An example of lettered categories would be:

- **Category A**: professional generative AI for cultural purposes, requiring an authorisation request from a government or an international body, accompanied by the detailed project of the film, audio, image, and so on. Not limited, since they enable certain technical feats, notably in cinema and music.
- **Category B**: professional generative AI for entertainment purposes, with slightly reduced capabilities and subject to an identity check and a licence to use, allowing digital content creators in particular to continue their activity. The licence could be withdrawn in the event of creating and distributing deepfakes.

- **Category C**: generative AI for personal use, with clearly reduced capabilities and subject to an identity check and a certificate of fitness to use, also making it possible to trace the origin of any deepfakes and thus to apply sanctions if needed.

Nevertheless, the ethical question arises: are we undermining freedom of expression? To what extent is that acceptable? Several surveys have already been conducted on public opinion regarding the uses of AI, their control by governments and the perception of their future in fields such as the improvement of working conditions, entertainment, health or politics. In particular, it is interesting to distinguish "expert" AI users (those who study it and use it for professional purposes) from the rest of the population. It has to be said that there are wide differences in perception between AI experts and the general population as to the social benefits of AI: in the United States, AI experts are far more enthusiastic than the others (McClain et al., 2025). Nevertheless, in the fields that concern us, such as elections, the two categories agree that the use of AI does not and will not improve the electoral process. As for regulation, again in the United States, the results are similar between experts and non-experts: 55 and 57 % of these categories respectively want more government control, and all of them are more worried about government laxity than about excessively strict control.

4.2.2 Hindering the distribution of deepfakes

A second option would again be to create an international framework for the non-distribution of deepfakes. But one may also think of national directives that would criminalise the relaying of false information, whether deliberate or not. This lever would in fact be twofold: on the one hand it would deter people from distributing unverified and therefore potentially false information; on the other hand it would encourage everyone to check the source and the truthfulness of the content they are about to share. This would however require a very precise legal framework in order to define the type of content to be banned, and great care will be needed so as not to fall into the trap of abusive and authoritarian censorship.

5 Educate

5.1 Improving media and information literacy

At this last level, the aim would be to raise awareness and to educate the population about the implications of viewing deepfake content. In reality, this recourse is also the pinnacle of protection: if no one, at the individual level, is any longer led astray as to the truthfulness of the information they receive, one may infer that only ill-intentioned individuals would be likely to distribute it, and therefore that legal sanctions would be better targeted and far less debatable. But how can this ideal be reached? One may consider integrating a deepfake component into media and information literacy (MIL) from the earliest age, typically from primary school onwards. This academic MIL will moreover have to be complemented by public awareness campaigns such as those that emerged from 1980 onwards to fight AIDS: posters, advertising slogans, publications, and so on.

5.2 Active and regular learning

Finally, the user must be put in the loop so as to become active in this cognitive struggle: for example, by periodically gating access to a social network with a "good practice" questionnaire on opening, which is repeated until a certain score is obtained and which resets each week or each month with questions of increasing difficulty.

In terms of scoring, one may also consider creating a quantification of the reliability level of internet users and displaying this score with each post, so that everyone can grant whatever credit they wish to a given post. A star-rating system, as on online commerce, accommodation or restaurant sites, would be a highly effective application: it would act both on the user, who must be careful about what they publish, and on the content consumer, who will pay increased attention to the rating of the profile in question. Another solution would be a repost justification form, along the lines of those that exist for reporting illegal content: it would force people to think about the meaning of their digital practices and would draw their attention to certain points of the image and/or the audio, with questions such as "do you find the person's gaze natural?", "are the movements coherent?", and so on.

Other answers to this problem are to be found on the side of social psychology. Indeed, behind the spread of any rumour there are human mechanisms that must be clearly identified in order to be fought. According to one study, some of the main factors in the success of rumours and in their rapid propagation are the lack of critical thinking and discernment, correlated with limited literacy and general knowledge, together with confirmation bias (Beauvais, 2022). Running large information campaigns about the sensitive topics of the moment can have an effect on some people by increasing their understanding, but not on others, notably because of confirmation bias. Another proposed approach would be to invert the recommendation algorithms of the distribution platforms. Social networks tend to promote content on the basis of what the consumer has already viewed, thereby pinning down the emotions they feel. By identifying precisely the emotional and sensational potential of a piece of information, one could manage to slow its propagation algorithmically, which would give *fact-checking* bodies enough time to rule on its truthfulness. Care will nonetheless be needed to ensure that information circulates at a reasonable speed, so as to guard against possible criticism of the *gatekeeping* kind. The criteria we propose for quantifying this emotional potential are the following: how recent the facts are, the level of interest of the population and the associated level of controversy.

6 Towards an offensive use of deepfakes?

6.1 A trend already under way

As seen above, deepfakes can prove very effective in influence and manipulation operations. This is why certain Western countries are contemplating their use. Thus the *United States Special Operations Command* (SOCOM), the

command centre of the American special forces, has already started working on a use of deepfakes. Members of the general staff are addressing the subject, together with psychologists and social media experts, in order to set up influence systems. They say they have developed tools that allow them to map the social media environment, to analyse trends and divergences of opinion in order potentially to influence this environment with their own tools.

Accordingly, in an unclassified document (USSOCOM, 2023), SOCOM announces that it has launched research that would "provide a next generation deepfake or similar technology in order to generate messages and influence operations via non-traditional channels in relevant environments against peer or near-peer adversaries". Here the intent to use deepfakes offensively is therefore openly acknowledged (Rempfer, 2019). In December 2024, *The Intercept* reported that SOCOM had persuaded X (Fang, 2022; Biddle, 2023), in violation of its own internal rules, to authorise a network of fake accounts that spread false information, notably one according to which the Iranian government was stealing the organs of Afghan civilians. Although this propaganda offensive did not use deepfakes, researchers found that Pentagon contractors had employed machine-learning-generated avatars in order to give these fake accounts a certain degree of realism. Likewise, in a document published by the *NATO Strategic Communications Center of Excellence* (STRATCOM) (Giles, 2019), NATO considers deepfakes a potentially central information warfare tool in hybrid conflicts, and wishes to learn to inject them into its own exercises. Although NATO has not published a categorical statement indicating that it commits never to use offensive deepfakes, its official documents place the emphasis on understanding, prevention and resilience in the face of these technologies, as part of protecting freedom of expression and democratic values, which suggests that it does not consider them legitimate tools of offensive influence (NATO, 2025).

To conclude, Western countries too are beginning to consider seriously the use of deepfakes as a more or less offensive instrument of information warfare.

6.2 Use on the battlefield

Alternatively, rather than an information warfare tool used at a distance as part of influence operations, could deepfakes not be used more directly in the field? Deepfakes would then be used as decoys. Thus, to support their offensive on Iran last June, the United States could have, in addition to their diversion strategy, made use of deepfakes to make it even more credible. Iranian intelligence had learned of the departure of American B2 aircraft towards the Pacific. This was in fact a decoy orchestrated by the American operations directorate, while the aircraft that were to carry out the bombing left later by the Atlantic (AFP, 2025). The United States could thus have posted, through well-concealed fake accounts, a faked video showing B2s carrying out an exercise in the Pacific, which would have led the Iranian services to believe that this was not a real attack.

7 Conclusion

Although the fight against deepfakes is only beginning, and despite the fact that this weapon evolves faster than the technical solutions available to defend against it, the war is not yet lost. There is nevertheless one highly sensitive dimension that we have only been able to touch on briefly here: ethics. Indeed, when it comes to regulating information or to using deepfakes offensively, a substantial amount of reflection remains to be carried out with respect to the implications for democracy, for human rights and for the legitimacy of Western states in the conduct of war. If in the past it has been possible for states such as France to be deployed on foreign theatres of operation without difficulty, that was partly thanks to the trust granted by host countries to the values and ideals France defends. What would happen if certain moral principles had to be given up in order to rebalance the global game of powers? How would the Nation then perceive its own country? And, from within as from without, who would its supporters be? These are all questions that would deserve an entire report of their own, but that need to be resolved so that Western democracies can fight their adversaries on an equal footing.

References

1. Abdulhamied, E. (2025). *Deepfake audio detection using feature-based and deep learning approaches: ANN vs ResNet50*. International Journal of Advanced Computer Science and Applications.
2. AFP. (2025). *Iran: ce qu'il faut savoir sur les frappes des États-Unis visant trois sites nucléaires*. Le Monde.
3. Barthélemy, L.-P. (2025). *Guerre en Ukraine et deepfakes: un outil discursif, stratégique et systémique*. Revue du Centre de recherche Analyse du discours.
4. Beauvais, C. (2022). *Fake news: why do we believe it?*
5. Bendiab, E. (2025). *Audio-deepfake: Generation methods, legitimate applications and the potential for misuse*.
6. Biddle, S. (2023). *U.S. Special Forces want to use deepfakes for psy-ops*. The Intercept.
7. Bonnet, F., Fidelin-Durand, G., Guinet, M., Morel, J. & Neveu, C. (2025). *Génération de deepfakes*. ENSTA.
8. Breland, A. (2019). *The bizarre and terrifying case of the "deepfake" video that helped bring an African nation to the brink*. Mother Jones.
9. Byman, D., Gao, F., Meserole, C. & Subrahmanian, V. S. (2023). *Deepfakes and international conflict*. Foreign Policy at Brookings.
10. Cinar, B. (2023). *Deepfakes in cyber warfare: threats, detection, techniques and countermeasures*. Asian Journal of Research in Computer Science.
11. Deromas-Lebocq, I. (2026). *Deepfakes Grok: la France saisit le procureur face à une explosion de contenus sexuels générés par IA*.
12. Desmarais, A. (2025). *Denmark fights back against deepfakes with copyright protection. What other laws exist in Europe?* Euronews.
13. Fang, L. (2022). *Twitter aided the Pentagon in its covert online propaganda campaign*. The Intercept.
14. Freeman, E. (2025). *The deepfake dilemma: can blockchain restore truth?* CoinGeek.
15. French Government. (2024). *Protect your image rights*.
16. Giles, K. (2019). *The role of deepfakes in malign influence campaigns*. NATO STRATCOM.
17. Johnson, A. (2019). *Report on a one-day expert workshop on understanding deepfakes and other forms of synthetic media in Subsaharan Africa*. Centre for Human Rights.
18. Kashif-Janjua, D. R. (2025). *Operation Sindoor: Indian media's black propaganda and fake news tactics in the 2025 India-Pakistan conflict*. Pakistan Social Sciences Review.
19. Kazic, T. (2024). *Digital propaganda and disinformation generated via artificial intelligence: the Israel/Palestine conflict and the fall of Bashar al-Assad in Syria*. Center for Strategic Prognosis.
20. Lassere, F. & Aleexeva, O. (2024). *Carte à la une. En mer de Chine méridionale, le jeu politique de l'interprétation du droit de la mer*.
21. Lokhande, M. (2024). *Artificial intelligence for detecting cyber attacks in deepfake and identity theft*. IEEE.
22. Majchrzak, A. (2023). *Russian disinformation and the use of images generated by artificial intelligence (deepfake) in the first year of the invasion of Ukraine*. University of Gdansk.
23. McClain, C. et al. (2025). *How the U.S. public and AI experts view artificial intelligence*. Pew Research Center.
24. NATO. (2025). *NATO's approach to counter information threats*.
25. Owasp. (2023). *Deepfakes: a growing cybersecurity concern*.
26. Patil, U. (2021). *An effective blockchain technique to resist against deepfake videos*.
27. Rashid, M. M. (2021). *Blockchain technology for combating deepfake and protect video/image integrity*. Journal of Korea Multimedia Society.
28. Rempfer, K. (2019). *SOCOM needs to step up its propaganda game, Pentagon deputy says*. Military Times.
29. Singh, K. (2025). *AI-generative deepfake content used as information war in the post-truth era*. International Journal of Advanced Mass Communication and Journalism.
30. Thomson, H. (2023). *A.I. muddies Israel-Hamas war in unexpected way*. The New York Times.
31. UN News. (2025). *Les armes à sous-munitions ont fait des centaines de morts et de blessés civils en Ukraine depuis 2022*.
32. United Nations Office for Disarmament Affairs. (2026).
33. USSOCOM. (2023). *Special operations forces acquisition, technology, and logistics directorate of science and technology (SOF AT&L-ST)*.